Summary of Current Model Verification Study

Goal

The goal of this round was to find a concrete fermionic model that can support the perturbative SDP improvement framework in the draft. In particular, we wanted to verify three points:

1. exactness of the reduced SDP at a reference Hamiltonian H_0;
2. first-order response matching between the reduced SDP and the physical ground state at H_0;
3. a nontrivial lower bound

$$\|\delta m_S(\theta)\| \geq c_{\mathrm{mis}}|\theta|$$

   for selected moments under the perturbative Hamiltonian $H(\theta) = H_0 + \theta V$.

We tested two spinful Hubbard-type candidates, both based on an isolated Hubbard-dimer reference point.

Common Setup And Notation

- Lattice: an open one-dimensional chain with L sites, L even.
- Filling: half filling, $N_e = L$ electrons.
- In this report, $D_r = (2r-1, 2r)$, $r = 1, \dots, L/2$, denotes the two-site unit that is coupled in the reference Hamiltonian H_0.
- In this report, $B_r = (2r, 2r+1)$, $r = 1, \dots, L/2-1$, denotes the inter-dimer bond on which the perturbation is applied.
- Operators:
  - $c_{i,\sigma}, c_{i,\sigma}^\dagger$: fermion annihilation/creation operators at site i with spin $\sigma \in \{\uparrow, \downarrow\}$.
  - $n_{i,\sigma} = c_{i,\sigma}^\dagger c_{i,\sigma}$, and $n_i = n_{i,\uparrow} + n_{i,\downarrow}$.
  - $D_i = n_{i,\uparrow} n_{i,\downarrow}$, the onsite double occupancy.
  - $K_{ij} = \sum_\sigma (c_{i,\sigma}^\dagger c_{j,\sigma} + c_{j,\sigma}^\dagger c_{i,\sigma})$, the bond kinetic observable.
  - $S_i^z = (n_{i,\uparrow} - n_{i,\downarrow})/2$, $S_i^+ = c_{i,\uparrow}^\dagger c_{i,\downarrow}$, and $S_i^- = c_{i,\downarrow}^\dagger c_{i,\uparrow}$.
  - $h.c.$ denotes the Hermitian conjugate.

The common reference Hamiltonian is the isolated-dimer spinful Hubbard model with $t_d = 1$:

$$H_0 = \sum_{r=1}^{L/2} \left[-K_{2r-1,2r} + U(D_{2r-1} + D_{2r}) \right].$$

For response matching, fields are coupled to the selected moments $M_1, \dots, M_s$ by

$$H(\theta, h) = H_0 + \theta V - \sum_{a=1}^{s} h_a M_a.$$

Reported Quantities

The quantities in this subsection are diagnostics used in this numerical report, not standard names from the Hubbard-model literature.

- Reference energy error: $|E_{\mathrm{SDP}}(H_0) - E_{\mathrm{phys}}(H_0)|$.
- Selected-moment error: $\|m_S^{\mathrm{SDP}}(0) - m_S^{\mathrm{phys}}(0)\|$.

- Response-matching error: $\|D_h m_S^{\mathrm{SDP}}(0) - D_h m_S^{\mathrm{phys}}(0)\|$, the norm of the difference between the SDP and physical first-order response matrices.
- Mismatch ratio: $\|\delta m_S(\theta)\|/|\theta|$, where $\delta m_S(\theta) = m_S^{\mathrm{SDP}}(\theta) - m_S^{\mathrm{phys}}(\theta)$. This ratio is used to estimate c_{mis}.
- Solver: all reported SDP values below were computed with MOSEK.

Terminology And References

- The terms "Hubbard model", "spinful Hubbard chain", "hopping", "onsite interaction", "half filling", and "double occupancy" follow the standard Hubbard-model terminology; see Hubbard's original paper and standard treatments of the one-dimensional Hubbard model [1, 2].
- The words "dimerized" and "alternating" refer to chains with alternating bond strengths; this terminology is standard in the dimerized or alternating Hubbard-chain literature [3, 4].
- The term "extended Hubbard model" refers here to adding an intersite density-density interaction to the onsite Hubbard interaction; this is the standard usage in the extended-Hubbard literature [5].
- "Response matrix" is used in the usual static linear-response sense, following the Kubo linear-response terminology [6].
- "Semidefinite programming" follows the standard convex-optimization terminology of Vandenberghe and Boyd [7].

Model 1: Standard Dimerized Spinful Hubbard Chain

Model. The first model is the standard alternating-hopping spinful Hubbard chain. It uses the common reference Hamiltonian H_0 above and the inter-dimer hopping perturbation

$$V_{\mathrm{hop}} = -\sum_{r=1}^{L/2-1} K_{2r,2r+1}.$$

The reported scan used $L = 4$, $U = 2$, $t_d = 1$, and $\theta = \pm 0.005, \pm 0.01$.

The selected moment families tested were:

- Bond kinetic moments within D_r: $M_r = K_{2r-1,2r}$.
- Double occupancy moments within D_r: $M_r = D_{2r-1} + D_{2r}$, and also the site-resolved versions D_{2r-1}, D_{2r}.
- Spin correlation moments within D_r: $S^z_{2r-1}S^z_{2r}$, and $\mathbf{S}_{2r-1} \cdot \mathbf{S}_{2r} = S^z_{2r-1}S^z_{2r} + \frac{1}{2}(S^+_{2r-1}S^-_{2r} + S^-_{2r-1}S^+_{2r})$.
- Onsite-pair hopping moments between the two sites of D_r: $P_r = c^\dagger_{2r-1,\uparrow}c^\dagger_{2r-1,\downarrow}c_{2r,\downarrow}c_{2r,\uparrow} + h.c.$.
- Singlet/triplet projectors on D_r: $\Pi_{s,r} = |s_r\rangle\langle s_r|$ and $\Pi_{t0,r} = |t_r^0\rangle\langle t_r^0|$, where $|s_r\rangle = (|\uparrow,\downarrow\rangle - |\downarrow,\uparrow\rangle)/\sqrt{2}$ and $|t_r^0\rangle = (|\uparrow,\downarrow\rangle + |\downarrow,\uparrow\rangle)/\sqrt{2}$ on dimer D_r.
- Bond kinetic moments on B_r: $M_r = K_{2r,2r+1}$.

Verification results.

1. **Reference exactness.** The reference SDP was solved at H_0. The following rows give representative moment choices from the scan.

Selected moments	Energy error	Moment error	Status
$K_{2r-1,2r}$	1.47×10^{-9}	9.37×10^{-8}	OPTIMAL
$D_{2r-1} + D_{2r}$	1.47×10^{-9}	4.74×10^{-8}	OPTIMAL
$K_{2r,2r+1}$	1.47×10^{-9}	4.65×10^{-16}	OPTIMAL

2. **First-order response matching.** Moments supported within D_r had small response-matching errors. The bond kinetic moment on B_r had a large response-matching error.

Selected moments	Support	Response error
$K_{2r-1,2r}$	within D_r	4.86×10^{-7}
$D_{2r-1} + D_{2r}$	within D_r	1.56×10^{-7}
$\mathbf{S}_{2r-1} \cdot \mathbf{S}_{2r}$	within D_r	1.44×10^{-7}
$\Pi_{s,r}$	within D_r	1.10×10^{-7}
$K_{2r,2r+1}$	on B_r	7.05

3. **Mismatch lower bound.** For moments supported within D_r, the ratio $\|\delta m_S(\theta)\|/|\theta|$ approximately doubled when $|\theta|$ doubled from 0.005 to 0.01. This indicates quadratic, not linear, scaling in the tested range. The bond kinetic moment on B_r had a stable linear ratio, but it failed the response-matching check above.

Selected moments	Support	Ratio at 0.005	Ratio at 0.01	Observation
$K_{2r-1,2r}$	within D_r	5.96×10^{-3}	1.17×10^{-2}	quadratic
$D_{2r-1} + D_{2r}$	within D_r	7.04×10^{-3}	1.40×10^{-2}	quadratic
$\mathbf{S}_{2r-1} \cdot \mathbf{S}_{2r}$	within D_r	6.36×10^{-3}	1.26×10^{-2}	quadratic
P_r	within D_r	5.42×10^{-3}	1.09×10^{-2}	quadratic
$K_{2r,2r+1}$	on B_r	4.59	4.59	linear ratio, response failed

Model 2: Density-Perturbed Extended Spinful Hubbard Dimer Chain

Model. The second model uses the same isolated Hubbard-dimer reference H_0, but replaces the hopping perturbation by an inter-dimer density-density perturbation:

$$V_{\text{dens}} = \sum_{r=1}^{L/2-1} n_{2r} n_{2r+1}.$$

The selected moments are the bond kinetic observables within D_r:

$$M_r = K_{2r-1,2r}, \qquad r = 1, \dots, L/2.$$

The reported checks used $U = 4$, $t_d = 1$, and $L = 4, 6$.

Verification results.

1. **Reference exactness.** At H_0, both the SDP energy and selected bond-kinetic moments matched the exact ground-state values to high precision.

System	Energy error	Moment error	Status
$L = 4, U = 4$	5.74×10^{-12}	1.01×10^{-7}	OPTIMAL
$L = 6, U = 4$	8.18×10^{-13}	1.03×10^{-9}	OPTIMAL

2. **First-order response matching.** The SDP and physical response matrices were compared by finite differences with field step $h = 0.005$.

System	Selected moments	Response error
$L = 4, U = 4$	$K_{2r-1,2r}$	3.04×10^{-5}
$L = 6, U = 4$	$K_{2r-1,2r}$	3.46×10^{-6}

3. **Mismatch lower bound.** The ratio $\|\delta m_S(\theta)\|/|\theta|$ stayed nearly constant over the tested θ-range, giving a numerical lower-bound constant.

System	Tested $\lvert\theta\rvert$	Ratio range	Conservative c_{mis}
$L = 4, U = 4$	$0.002, 0.005, 0.01, 0.02$	0.124–0.125	0.12
$L = 6, U = 4$	$0.005, 0.01$	0.215–0.216	0.21

Observed Status

For the standard hopping perturbation, the reference exactness error was small, the response check for moments supported within D_r passed numerically, and the corresponding mismatch was quadratic in θ in the tested cases.

For the density-density perturbation, the tested $L = 4$ and $L = 6$ cases both gave small reference errors, small response-matching errors, and a linear mismatch ratio bounded away from zero on the tested θ-range.

References

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